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RULE · Quadratic equation (kvadrat tenglama)

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$

Up to two roots – find and report both.

RULE · Factoring when $a = 1$

Find two numbers that multiply to c and add to b.

$$x^2 + 5x + 6 = 0 \implies (x + 2)(x + 3) = 0 \implies x = -2 \text{ or } -3$$

RULE · Factor out the GCF first

If every term shares a common factor, pull it out before factoring the rest.

$$2x^2 + 8x + 6 = 0 \implies 2(x + 1)(x + 3) = 0 \implies x = -1 \text{ or } -3$$

RULES SHEET – 2/2

RULE · Zero-product property

Works only when the product equals exactly 0. If it doesn't, rearrange everything to one side = 0 first.

$(x-3)(x+1)=5$

$x^2-2x-3=5 \implies x^2-2x-8=0 \implies (x-4)(x+2)=0$

$x=4$ or $x=-2$

A question often asks for the sum or product of the roots, not the roots themselves – e.g. sum $=4+(-2)=2$.